Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 30

Recursive Sequences and Strong Induction

Covers Chartrand & Zhang, Sections 4.3 and 4.4

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Ordinary or strong?

Ask: **how far back does the inductive step need to reach?**

- Only to $k \Rightarrow$ ordinary induction.
- To $k-1$, $k-2$, or somewhere unpredictable $\Rightarrow$ strong induction.

If the step reaches back j places, you need j consecutive base cases.

Three mistakes to expect

1. **Using a recurrence where it does not apply.** Check the "for $n \geq \ldots$" condition.
2. **Too few base cases.** If the step uses $k-3$, you need four starting values.
3. **Conjecturing from too few terms.** Compute at least four or five before guessing.

Part II: Practice Problems

Section A — Recursive sequences

1. Let $a_1 = 1$ and $a_n = a_{n-1} + 2n - 1$ for $n \geq 2$.

(a) Compute a_2 through a_5 and conjecture a formula.

(b) Prove your formula.

Section B — Strong induction

2. Let $a_1 = 1$, $a_2 = 2$, and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$. Prove that $a_n < 2^n$ for every positive integer n.

3. Prove that every amount of postage of 8 cents or more can be made using only 3-cent and 5-cent stamps.

4. Prove that every integer $n \geq 2$ is either prime or a product of primes.

Section C — Writing

5. In five to eight complete sentences, explain why a recursive function can be proved correct by strong induction. Describe what plays the role of the base step and the inductive hypothesis, and say what bug corresponds to a missing base case.